

IN-LAB EXERCISE:

OBJECTIVE:

To design and develop interactive dashboards in Tableau and Power BI for dynamic data exploration, enabling users to analyze, filter, and drill down into data for deeper insights.

PROCEDURE:

1. Data Collection:

- Import datasets from sources such as Excel, CSV, SQL Server, or online APIs.
- Dataset: *Retail Sales Dataset* with fields such as Order Date, Region, Product Category, Sales, Profit, Quantity.

2. Data Preparation and Transformation:

- Ensure proper date formats for time-based analysis.
- Handle missing or incorrect values.
- Create calculated fields (e.g., Profit Margin %, Sales Growth %, Year-to-Date Sales).

3. Data Aggregation and Hierarchies:

- Create hierarchies (e.g., Year → Quarter → Month → Day).
- Aggregate data by region, product category, or customer segment.

4. Designing Interactive Elements:

- **Filters & Slicers:** Allow filtering by product category, date range, or region.
- **Parameters:** Let users switch between KPIs (e.g., Sales vs. Profit).
- **Drill-down:** Enable users to explore data from summary to detail.
- **Tooltips:** Add contextual details when hovering over data points.

5. Dashboard Development:

- **Tableau:** Use interactive filters, parameter controls, and highlight actions.
- **Power BI:** Use slicers, bookmarks, and drill-through pages.
- Add KPI cards for key metrics like *Total Sales*, *Total Profit*, *Top Selling Product*.
- Use maps for regional sales analysis, line charts for trends, and bar charts for category comparisons.

6. Testing Interactivity:

- Ensure slicers and filters update all relevant visuals dynamically.
- Test drill-down and drill-through functionality for accuracy.

7. Publishing and Sharing:

1. How did interactivity improve the analysis experience in your dashboard?
2. Which interactive feature (filters, drill-down, parameters, etc.) was most useful and why?
3. What challenges did you face while implementing drill-down or dynamic filters?
4. How do tooltips contribute to better dashboard storytelling?
5. What additional features would you add to make the dashboard more insightful?

RESULT:

The retail sales dataset was successfully transformed, aggregated, and visualized into interactive dashboards in Tableau and Power BI. Filters, slicers, drill-downs, and parameters enabled dynamic exploration of sales and profit trends by region, product category, and time period. The resulting dashboards allowed for quick insights and deeper analysis, enhancing decision-making capabilities.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		